

Interim Report

Github Repository : <https://github.com/q62ur/CS4403-Term-Project>

1. A summary of the data you have collected. What format is it in? How many data instances do you have?

We collected four speech-based text datasets for this project.

The first dataset contains TED Talk transcripts, which represent longer spoken presentations. The second dataset is the Mozilla Common Voice dataset which contains short spoken sentences from many speakers. The third dataset is a synthetic speech dataset generated for this project to resemble natural spoken language. In addition, a TED metadata dataset containing information about the talks was also included.

The TED transcripts dataset (`transcripts.csv`) and the TED metadata dataset (`ted_main.csv`) were obtained from Kaggle. The Mozilla Common Voice dataset (`cv-valid-train.csv`) was obtained from the Mozilla Common Voice project. The synthetic speech dataset (`synthetic_speech_dataset_2000.csv`) was generated using ChatGPT.

All datasets are stored in CSV format and contain a column representing the text of the speech or sentence.

The combined datasets contain a total of 202,793 data instances.

Datasets	Instances
TED transcripts	2,467
Mozilla Common Voice	195,776
Synthetic Speech	2,000
TED Metadata	2550

2. Details of the progress you have made towards your project goal. Please note that a report on data collection is not sufficient for your interim report. You need to be working on the data at this point in time.

First, all datasets were successfully loaded and explored to understand their structure and relevant columns. The text columns containing speech content were extracted from each dataset.

Next, the datasets were combined into a single dataset with a label indicating the source of each sentence. This allows us to analyze and compare patterns across real and synthetic speech data.

The dataset was then cleaned and preprocessed. The cleaning process included:

- Removing missing values
- Removing duplicate sentences
- Converting text to lowercase
- Removing punctuation and special characters
- Removing sentences with fewer than three words

After cleaning, the dataset was saved as `cleaned_speech_dataset.csv` for further analysis.

We also performed exploratory data analysis, including:

- Visualizing the number of samples from each dataset
- Analyzing text length distributions
- Identifying the most important words using TF-IDF

We also implemented classification models to predict which dataset each sentence belongs to. The models were trained using TF-IDF features extracted from the speech text.

3. What data mining strategies have you explored so far, and with what results?

First, we applied TF-IDF to convert the text data into numerical features. This allowed the machine learning models to process the textual transcripts by representing each document as a vector of important terms. We also explored the most frequent TF-IDF terms for each source to understand which words were most characteristic of each dataset.

After preprocessing, we trained three classification models:

- Naive Bayes
- Logistic Regression
- Support Vector Machine

The results were as follows:

- **Naive Bayes**

- Accuracy: 95.9%
- Performance was good overall, but it performed slightly worse on the TED class compared to the others.

- **Logistic Regression**

- Accuracy: 99.7%
- Very high precision and recall across all classes.

- **Support Vector Machine (SVM)**

- Accuracy: 99.86%
- This produced the best performance, with nearly perfect classification.

Confusion matrices were also generated to visualize how well each model predicted the three sources.

The results showed that Logistic Regression and SVM could almost perfectly identify which dataset each text came from.

Overall, SVM performed the best.

4. What are your proposed strategies for other or further approaches?

More advanced text preprocessing techniques may be applied, such as removing stopwords or using stemming or lemmatization.

We may also experiment with different feature extraction techniques, such as:

- N-grams
- Word embeddings
- More advanced text representations

Additional models could be tested, such as:

- Random Forest
- Gradient Boosting
- Neural network models

5. A summary of what is left to do to complete your project and your assessment of how you will do so within the remaining time.

There are still several components left before the final report. The remaining work primarily focuses on finalizing visualizations, interpreting the results and organizing the findings into a clear final report. Additional analysis may also be performed to further compare the datasets and evaluate model performance.

In the first week, we plan to finalize visualizations of the dataset and model results. In the second week, we will experiment with additional preprocessing or feature extraction methods and then further analyze the model performance. In the third week, we will compare the results of different models and interpret the findings. In the final week, we will complete the final report and organize our results.

Since the main steps such as data collection, cleaning, feature extraction, and initial model implementation have already been completed, we believe the remaining work can be finished within the available time through coordinated effort among the team members.

6. For groups: Provide detail of each of the group members' contributions so far.

Tasks	Member
Idea Generation	Fatima
Collecting datasets	Chanchal
Data Loading, Cleaning, Combining	Chanchal
Feature Extraction	Medha
Classification Models	Medha
Visualization	Fatima
Model Analysis	Medha